

Impact of Input Membership Function Design on Fuzzy-Adaptive Particle Swarm Optimization: An Empirical Sensitivity Analysis

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Abstract—This study investigates the impact of input membership function design on the performance of a fuzzy-controlled Particle Swarm Optimization (PSO). While existing fuzzy PSO research treats input partitions as fixed design elements, this work isolates the effect of varying the linguistic representation of iteration progress and population diversity. Four input set geometries crossed with two granularity levels are evaluated on six benchmark functions under a constant output set and rule base. Experimental results demonstrate that the configuration of input fuzzy sets significantly influences the swarm's ability to escape local optima in complex multimodal landscapes, with moderate-overlap partitions and three-label granularity yielding the most robust performance.

Index Terms—Particle Swarm Optimization, fuzzy logic, inertia weight, membership functions, linguistic granularity, adaptive parameter control

I. INTRODUCTION

Particle Swarm Optimization (PSO) is a population-based metaheuristic whose performance is strongly influenced by the balance between global exploration and local exploitation [1]. The inertia weight plays a key role in regulating this balance, motivating numerous adaptive strategies aimed at improving convergence behavior and robustness [2].

Fuzzy control systems have been widely employed to adapt PSO parameters online using qualitative descriptions of the search state, providing a flexible alternative to deterministic or iteration-based schedules [3]. However, most fuzzy-controlled PSO variants rely on fixed linguistic partitions and predefined membership functions selected a priori, typically without explicit analysis of their semantic design.

As a result, the impact of linguistic granularity and membership function configuration remains insufficiently explored. In particular, it is unclear to what extent different fuzzy linguistic schemes, even when sharing identical control logic,

influence optimization behavior across problems with different landscape characteristics.

This paper presents an experimental analysis of fuzzy-controlled PSO variants that differ exclusively in linguistic granularity and membership function design. Rather than proposing a new optimization algorithm, the study isolates and evaluates the effect of alternative fuzzy semantic configurations under identical control logic, assessing their impact on performance and robustness across representative benchmark functions. The results position fuzzy scheme design as a relevant, yet often overlooked, experimental dimension in adaptive metaheuristic optimization.

The remainder of this paper is organized as follows. Section II reviews relevant work on fuzzy-adaptive parameter control in PSO and related optimization frameworks. Section III describes the fuzzy-controlled PSO framework and the linguistic configurations evaluated in this study. Section IV details the benchmark functions, parameter settings, and experimental protocol. The experimental results and their discussion are presented in Section V. Finally, Section VI summarizes the main findings and outlines directions for future research.

II. RELATED WORK

This section reviews two bodies of literature that underpin the present study: adaptive inertia weight strategies in PSO, with emphasis on fuzzy control approaches, and the influence of membership function design and linguistic granularity on fuzzy system performance.

A. Adaptive Inertia Weight Control in PSO

Since the introduction of the inertia weight by Shi and Eberhart [2], numerous strategies have been proposed to

replace the original linear decreasing schedule with state-dependent adaptation. Chatterjee and Siarry [4] introduced nonlinear decreasing functions that accelerate convergence in later iterations, while Nickabadi et al. [5] designed a feedback-based mechanism that adjusts w according to the swarm's success ratio, coupling the control signal to actual search progress rather than to a predefined temporal schedule.

Fuzzy logic provides a qualitative alternative by encoding adaptation knowledge as linguistic rules. Shi and Eberhart [3] first demonstrated that a Mamdani-type controller responding to swarm performance indicators can outperform static schedules, establishing a paradigm in which the inertia weight is governed by linguistic inference rather than an explicit function of the iteration count. Valdez et al. [6] extended this approach with co-evolutionary strategies and showed consistent advantages over classical PSO across diverse benchmarks. Nobile et al. [7] proposed a self-tuning mechanism that simultaneously adjusts multiple PSO parameters at the particle level, while Xia et al. [8] introduced a multiple-input multiple-output fuzzy controller for adapting learning coefficients and elite selection. Olivas et al. [9] showed that interval type-2 controllers can handle additional uncertainty from noisy landscapes, and Roy et al. [10] recently proposed evolving fuzzy policies that adapt up to seven hyperparameters, representing the most comprehensive fuzzy control architecture for PSO reported to date.

A common feature across all these contributions is that the input membership functions—defining how diversity, fitness progress, or other indicators are fuzzified—are selected a priori and their geometric configuration is not examined as an experimental factor. Research efforts have focused on rule base design, output set selection, or the number of controlled parameters, treating the input partition as a fixed design element.

B. Membership Function Design and Linguistic Granularity

Evidence from the broader fuzzy systems literature suggests that the input partition is far from a neutral design choice. Olivas et al. [9] demonstrated that different membership function shapes produce statistically significant performance differences even when the rule base remains unchanged, indicating that membership functions carry semantic information that modulates inference quality. Metaheuristic-based tuning of membership function parameters has confirmed this sensitivity across a range of applications [11], [12].

At the partition level, Cordon et al. [13] established guidelines for obtaining uniform fuzzy partitions with adequate granularity, showing that properties such as overlap degree and label spacing are constrained by consistency requirements. Guillaume and Charnomordic [14] proposed methods for generating interpretable partitions from data while preserving semantic coherence. These results establish that the geometric configuration of membership functions—support width, overlap, and boundary treatment—directly influences both the interpretability and the control behavior of the fuzzy system.

The number of linguistic labels used to partition the input space, referred to as linguistic granularity, constitutes a second design dimension. Castiello et al. [15] showed that adaptive granularity can improve discrimination in classification tasks without sacrificing interpretability. In optimization contexts, Huang et al. [16] and Zhang et al. [17] demonstrated that the granularity of linguistic scales affects convergence behavior and solution quality in PSO-related decision-making problems.

Despite this evidence, the specific influence of input membership function geometry and linguistic granularity on fuzzy-controlled PSO remains unexplored. Existing fuzzy PSO studies adopt a single partition—typically three uniformly spaced triangular labels—without justifying this choice or evaluating alternatives. The present work addresses this gap through a factorial design in which four input set geometries, differing in overlap degree and boundary treatment, are crossed with two granularity levels (3 and 5 labels) under a constant output set and rule base. This design isolates the input partition as the sole experimental variable, enabling direct assessment of how fuzzification geometry affects optimization performance across problems with distinct landscape characteristics.

III. FUZZY-CONTROLLED PSO FRAMEWORK

This section describes the fuzzy-controlled Particle Swarm Optimization (PSO) framework employed in the experimental evaluation. The standard PSO algorithm is extended by incorporating a swarm-level fuzzy control system that dynamically adapts the inertia weight during the optimization process. The central research question addressed in this work is how the geometric configuration of the *input* membership functions affects controller behavior and optimization performance. Accordingly, the output fuzzy set and rule base are held constant across all variants, isolating the input partition as the sole experimental variable.

A. Baseline Particle Swarm Optimization

The baseline corresponds to the canonical PSO formulation, where particle motion is governed by inertia, cognitive attraction toward the personal best, and social attraction toward the global best position. Particle dynamics are defined by

$$v_i(t+1) = w \cdot v_i(t) + c_1 r_1 (p_i - x_i(t)) + c_2 r_2 (g - x_i(t)), \quad (1)$$

$$x_i(t+1) = x_i(t) + v_i(t+1), \quad (2)$$

where p_i and g denote the personal and global best positions, and $r_1, r_2 \in [0, 1]$ are uniformly distributed random variables.

In the baseline configuration, the inertia weight follows a deterministic linear decreasing schedule from 0.9 to 0.1 over the optimization horizon [18]. The acceleration coefficients are set to $c_1 = c_2 = 2.0$ for all configurations. This serves as the reference for assessing the effect of fuzzy-controlled inertia adaptation.

B. Fuzzy Control System for Inertia Weight Adaptation

In the fuzzy-controlled variants, a Mamdani-type fuzzy inference system (FIS) is embedded into the PSO loop and evaluated once per iteration at the swarm level. At iteration t , the controller outputs an inertia weight w_t based on two normalized indicators: iteration progress and population diversity.

Iteration progress is defined as

$$t_{\text{progress}} = \frac{t}{T_{\text{max}}}, \quad (3)$$

where T_{max} is the maximum number of iterations. Population diversity measures the normalized spatial dispersion of the swarm:

$$D(t) = \frac{1}{d} \sum_{k=1}^d \frac{\max_i x_{i,k}(t) - \min_i x_{i,k}(t)}{U_k - L_k}, \quad (4)$$

where d is the problem dimensionality and $[L_k, U_k]$ are the bounds of dimension k .

The fuzzy controller maps these inputs to an inertia weight constrained to $[w_{\min}, w_{\max}]$. Defuzzification is performed using the centroid method:

$$w_t = \frac{\int z \mu_{\text{agg}}(z) dz}{\int \mu_{\text{agg}}(z) dz}, \quad (5)$$

where $\mu_{\text{agg}}(z)$ denotes the aggregated output membership function. Fig. 1 illustrates the integration of the fuzzy controller within the PSO iteration loop.

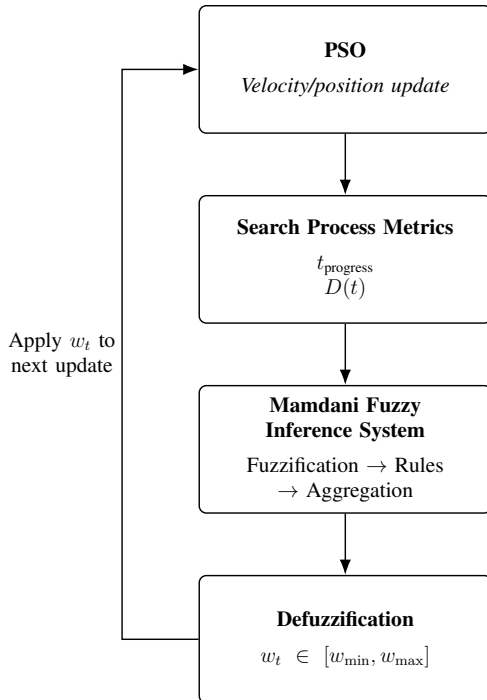


Fig. 1. Architecture of the Fuzzy-Controlled PSO Framework

C. Input Membership Function Configurations

The central experimental factor is the geometric configuration of the input membership functions that partition the diversity ratio and iteration progress domains. While prior work on fuzzy-controlled PSO has explored diverse rule bases, multi-parameter tuning, and alternative controller architectures [3], [7], [8], the sensitivity of the controller to input partition geometry has received limited attention. This work addresses that gap by evaluating four input set configurations, each employing triangular membership functions with different overlap and boundary characteristics.

All four configurations use the same linguistic labels—low, medium, and high for diversity; early, mid, and late for iteration progress—and are applied symmetrically to both input variables. This ensures that differences in controller behavior arise exclusively from the geometric properties of the partitions, not from changes in the semantic structure.

- **I1 (Standard):** Uniformly distributed triangular functions with moderate overlap, providing balanced coverage of the input domain.
- **I2 (Narrow):** Reduced support width and zero overlap between adjacent labels, producing sharper transitions and a more selective control response.
- **I3 (Wide):** Increased support width and substantial overlap, yielding smoother interpolation and gentler inertia weight adaptation.
- **I4 (Shoulder):** Shoulder functions at the extremes with a standard triangular function at the center. Extreme labels maintain full activation ($\mu = 1$) near domain boundaries, assigning maximum confidence to a single rule at extreme input values.

Additionally, two levels of linguistic granularity are evaluated: a three-label partition (low, medium, high) and a five-label partition (very_low, low, medium, high, very_high). In the five-label case, each input set extends its geometric principles—overlap degree, boundary treatment—to five uniformly spaced labels. Crossing four input geometries with two granularity levels yields eight fuzzy-controlled PSO variants.

Tables I and II detail the triangular parameters (a, b, c) for each configuration at both granularity levels. The same partition is applied symmetrically to both input variables.

D. Output Fuzzy Set Configuration

The output variable is the inertia weight w , defined over the normalized universe of discourse $[0, 1]$ and subsequently scaled to $[w_{\min}, w_{\max}]$. A single output set configuration (Set A) is used across all variants, ensuring that the output membership functions do not confound the effect of the input partition under study.

Set A employs uniformly distributed triangular membership functions with symmetric overlap. For the three-label case, the output labels are low (0.00, 0.25, 0.5), medium (0.25, 0.50, 0.75), and high (0.50, 0.75, 1.0). The five-label extension subdivides the same domain into very_low, low, medium, high, and very_high with equal spacing.

TABLE I

THREE-LABEL TRIANGULAR MF PARAMETERS (a, b, c) FOR EACH INPUT SET. IN I4, EXTREME LABELS USE SHOULDER FUNCTIONS WHERE $a=b$ OR $b=c$.

Label	I1 (Standard)	I2 (Narrow)	I3 (Wide)	I4 (Shoulder)
Low/Early	(0.0, 0.2, 0.4)	(0.0, 0.16, 0.33)	(0.0, 0.25, 0.5)	(0.0, 0.0, 0.35)
Medium/Mid	(0.3, 0.5, 0.7)	(0.33, 0.5, 0.66)	(0.25, 0.5, 0.75)	(0.2, 0.5, 0.8)
High/Late	(0.6, 0.8, 1.0)	(0.66, 0.84, 1.0)	(0.5, 0.75, 1.0)	(0.65, 1.0, 1.0)

TABLE II

FIVE-LABEL TRIANGULAR MF PARAMETERS (a, b, c) FOR EACH INPUT SET. I4 RETAINS SHOULDER FUNCTIONS AT EXTREMES; I2 ACHIEVES ZERO OVERLAP BETWEEN ADJACENT LABELS.

Label	I1 (Standard)	I2 (Narrow)	I3 (Wide)	I4 (Shoulder)
Very Low/Very Early	(0.0, 0.15, 0.3)	(0.0, 0.1, 0.2)	(0.0, 0.15, 0.35)	(0.0, 0.0, 0.25)
Low/Early	(0.2, 0.35, 0.5)	(0.2, 0.3, 0.4)	(0.15, 0.3, 0.5)	(0.15, 0.3, 0.45)
Medium/Mid	(0.35, 0.5, 0.65)	(0.4, 0.5, 0.6)	(0.3, 0.5, 0.7)	(0.35, 0.5, 0.65)
High/Late	(0.5, 0.65, 0.8)	(0.6, 0.7, 0.8)	(0.5, 0.7, 0.85)	(0.55, 0.7, 0.85)
Very High/Very Late	(0.7, 0.85, 1.0)	(0.8, 0.9, 1.0)	(0.65, 0.85, 1.0)	(0.75, 1.0, 1.0)

E. Fuzzy Rule Base

The rule base defines the qualitative mapping between iteration progress, population diversity, and inertia weight. The underlying control strategy follows a standard exploration-to-exploitation transition: high inertia is promoted when diversity is low or the search is in early stages, while low inertia is assigned in later stages with sufficient diversity.

The same rule semantics are preserved across all input set configurations and both granularity levels. Tables III and IV present the complete rule bases.

TABLE III

FUZZY RULE BASE FOR INERTIA WEIGHT ADAPTATION (3 LABELS, 9 RULES).

Rule	Progress	Diversity	Output w
1	early	low	high
2	early	medium	high
3	early	high	high
4	mid	low	low
5	mid	medium	medium
6	mid	high	high
7	late	low	low
8	late	medium	low
9	late	high	medium

TABLE IV

FUZZY RULE BASE FOR INERTIA WEIGHT ADAPTATION (5 LABELS, 25 RULES).

Rule	Progress	Diversity	Output w
1	very early	very low	very high
2	very early	low	very high
3	very early	medium	high
4	very early	high	high
5	very early	very high	medium
6	early	very low	very high
7	early	low	high
8	early	medium	high
9	early	high	medium
10	early	very high	medium
11	mid	very low	high
12	mid	low	high
13	mid	medium	medium
14	mid	high	medium
15	mid	very high	low
16	late	very low	high
17	late	low	medium
18	late	medium	low
19	late	high	low
20	late	very high	very low
21	very late	very low	medium
22	very late	low	low
23	very late	medium	very low
24	very late	high	very low
25	very late	very high	very low

IV. EXPERIMENTAL SETUP

A. Benchmark Functions

The experimental evaluation uses six classical benchmark functions representing two structurally distinct groups: high-dimensional problems with regular landscapes and low-dimensional problems with rugged, deceptive landscapes.

The first group includes three functions evaluated in $d = 100$ dimensions. The Sphere function (F1) serves as a unimodal baseline with a smooth, convex landscape. Rosenbrock's function (F5) provides a unimodal challenge characterized by a narrow, curved valley where convergence is highly

sensitive to search direction. Griewank's function (F11) introduces regular multimodality through oscillatory components superimposed on a quadratic base.

The second group comprises three variants of the Shekel function in a four-dimensional space ($d = 4$), differing in the number of components m : F21 ($m=5$), F22 ($m=7$), and F23 ($m=10$). The Shekel family is characterized by closely spaced deep optima, making it particularly challenging for population-based methods prone to premature convergence.

All functions were implemented using the `Opfunu` li-

brary [19], ensuring reproducibility. Table V summarizes the selected benchmarks.

B. Parameter Settings and Evaluation Protocol

All algorithmic variants were evaluated under identical conditions. The population size was fixed at 50 particles and the maximum number of iterations was set to 500. No additional stopping criteria were applied.

Each algorithm–function combination was executed over 31 independent runs. A shared seeding strategy was adopted: all algorithms used the same random seed for the k -th run, with the initial seed set to 42 and incremented by one for each subsequent run, yielding seeds $\{42, 43, \dots, 72\}$.

The baseline PSO with linearly decreasing inertia weight was evaluated alongside eight fuzzy-controlled variants (PSO–FCS). These variants result from the full factorial combination of the four input set configurations described in Section III-C (I1–I4) and two granularity levels (3 and 5 labels), with the output fuzzy set (A) and rule base held constant. Table VI summarizes the nine configurations, totaling $9 \times 6 \times 31 = 1,674$ independent experiments.

Performance was assessed using the final best fitness value and the relative gap to the known global optimum, defined as

$$\text{Gap}(\%) = \frac{|f_{\text{best}} - f_{\text{opt}}|}{|f_{\text{opt}}|} \times 100. \quad (6)$$

For functions with $f_{\text{opt}} = 0$ (F1, F5, F11), performance is reported directly as the absolute best fitness value, since the relative gap is undefined.

Robustness was evaluated via the mean and standard deviation of final fitness values across the 31 runs. Statistical significance was assessed using the Wilcoxon signed-rank test at $\alpha = 0.05$, which exploits the paired design enabled by shared random seeds. Computational cost was measured using wall-clock execution time.

All experiments were conducted on a 64-bit system with an Intel Core i7 CPU (2.8 GHz) and 16 GB of RAM. The algorithms were implemented in Python 3.11.9 and executed sequentially.

V. RESULTS AND DISCUSSION

This section presents the experimental results organized in three parts: the overall comparison between the baseline PSO and the fuzzy-controlled variants, the effect of input set geometry on controller performance, and the influence of linguistic granularity.

A. Overall Performance: PSO vs PSO–FCS

Table VII summarizes the comparison between the baseline PSO and the best-performing PSO–FCS variant for each benchmark function. The best fuzzy configuration per function was selected by lowest mean fitness (for minimization problems F1, F5, F11) or highest mean fitness (for maximization problems F21–F23) across 31 runs.

A stark dichotomy emerges between unimodal and multimodal landscapes. On F1, F5, and F11 the baseline PSO

achieves fitness values that are orders of magnitude better than any fuzzy variant ($p < 0.001$, Wilcoxon signed-rank test), while incurring 20–28% less computation time. This result is expected: unimodal landscapes with a single basin of attraction do not benefit from diversity-driven inertia adaptation. The fuzzy controller interprets the natural convergence of the swarm as a signal to increase exploration, which is counter-productive in a landscape where convergence is desirable from the outset.

On the Shekel functions (F21–F23), the pattern reverses. The best PSO–FCS variant improves mean fitness by 6.5–22.9% relative to the baseline, while reducing standard deviation by 57–63%. The computational overhead is marginal (1.0–5.3%). These results indicate that the fuzzy controller successfully detects premature convergence toward local optima and reactivates exploration through increased inertia. The substantial reduction in variability suggests that the controller stabilizes the search process, yielding more consistent outcomes across independent runs.

However, the statistical significance of the multimodal results is limited. Only F21 achieves $p < 0.05$ with the Wilcoxon test ($p = 0.041$), while F22 and F23 do not reach significance at the 5% level. This is attributable to the inherent variance of the Shekel family: with closely spaced deep optima, even small perturbations in initial conditions lead to different attractors, inflating within-group variance and reducing statistical power.

Fig. 2 visualizes the fitness distributions for all six functions, confirming the dichotomy between function groups. For the Shekel functions, the fuzzy variants exhibit tighter interquartile ranges and fewer extreme outliers compared to the baseline.

The fitness–robustness tradeoff is further illustrated in Fig. 3, which plots mean fitness against standard deviation for all nine algorithm configurations on each benchmark function. On the Shekel functions, the PSO baseline (star marker) occupies the high-variance region of the scatter, while the FCS variants cluster toward lower standard deviations. On unimodal functions, all FCS configurations collapse into a distant cluster far from the PSO baseline, visually confirming their inability to compete on these landscapes.

B. Effect of Input Set Geometry

The central question of this work is whether the geometric configuration of the input membership functions affects the performance of the fuzzy controller. Table VIII reports the mean rank of each input set across all six benchmark functions, computed separately for each granularity level. A rank of 1 indicates the best-performing configuration on a given function; lower average rank implies more robust performance across the benchmark suite.

I1 (Standard), with uniformly spaced triangular functions and moderate overlap, achieves the best average rank (2.83) in the three-label configuration. I2 (Narrow) ranks second, suggesting that sharper transitions between linguistic labels can be beneficial, particularly on the Shekel functions where rapid reaction to diversity changes is advantageous. I3 (Wide) is consistently the worst-performing configuration at both

TABLE V
BENCHMARK FUNCTIONS USED IN THE EXPERIMENTAL EVALUATION.

Function	Type	Dim.	Bounds	Global Opt.
F1 (Sphere)	Unimodal	100	$[-100, 100]^d$	0.0
F5 (Rosenbrock)	Unimodal	100	$[-30, 30]^d$	0.0
F11 (Griewank)	Multimodal	100	$[-600, 600]^d$	0.0
F21 (Shekel, $m=5$)	Multimodal	4	$[0, 10]^d$	-10.1532
F22 (Shekel, $m=7$)	Multimodal	4	$[0, 10]^d$	-10.4029
F23 (Shekel, $m=10$)	Multimodal	4	$[0, 10]^d$	-10.5363

TABLE VI
ALGORITHM CONFIGURATIONS EVALUATED. ALL PSO-FCS VARIANTS SHARE THE SAME RULE BASE AND OUTPUT SET; ONLY INPUT MEMBERSHIP FUNCTIONS AND GRANULARITY VARY.

Algorithm	w Control	Output Set	Labels	Input Set
PSO (baseline)	Linear decrease	—	—	—
PSO-FCS-I1-3L	Fuzzy (FCS)	A	3	I1 (Standard)
PSO-FCS-I2-3L	Fuzzy (FCS)	A	3	I2 (Narrow)
PSO-FCS-I3-3L	Fuzzy (FCS)	A	3	I3 (Wide)
PSO-FCS-I4-3L	Fuzzy (FCS)	A	3	I4 (Shoulder)
PSO-FCS-I1-5L	Fuzzy (FCS)	A	5	I1 (Standard)
PSO-FCS-I2-5L	Fuzzy (FCS)	A	5	I2 (Narrow)
PSO-FCS-I3-5L	Fuzzy (FCS)	A	5	I3 (Wide)
PSO-FCS-I4-5L	Fuzzy (FCS)	A	5	I4 (Shoulder)

PSO vs PSO_FCS (all configs pooled): Fitness Distribution

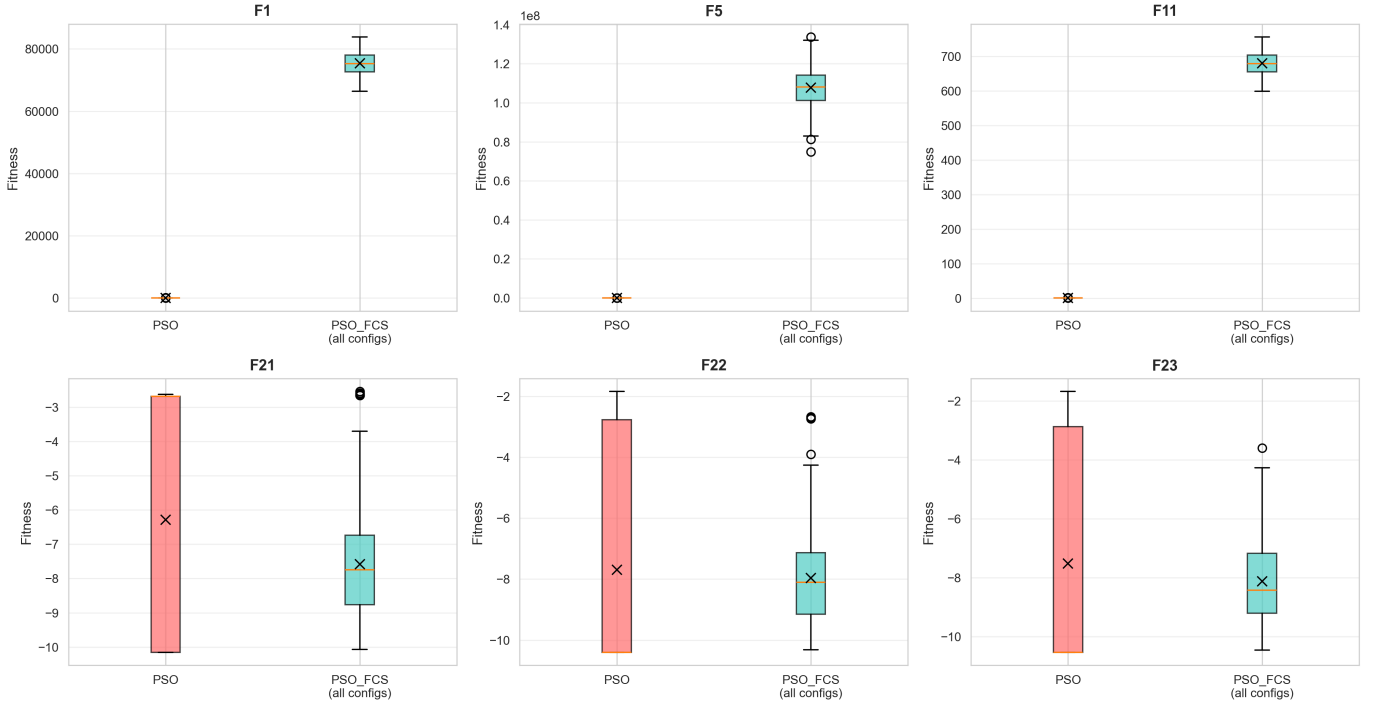


Fig. 2. Fitness distributions (31 runs) for PSO and the best PSO-FCS variant per function. On unimodal functions (F1, F5, F11), PSO dominates. On multimodal functions (F21–F23), the fuzzy-controlled variant achieves better medians and reduced dispersion.

granularity levels, indicating that excessive overlap dilutes the control signal and produces an overly smooth inertia adaptation that fails to distinguish between meaningfully different input states.

Fig. 4 presents the fitness heatmap for the multimodal functions, where the effect of input set geometry is most pronounced. I1 and I2 consistently achieve the darkest (best) cells, while I3 and I4 show lighter (worse) values, particularly

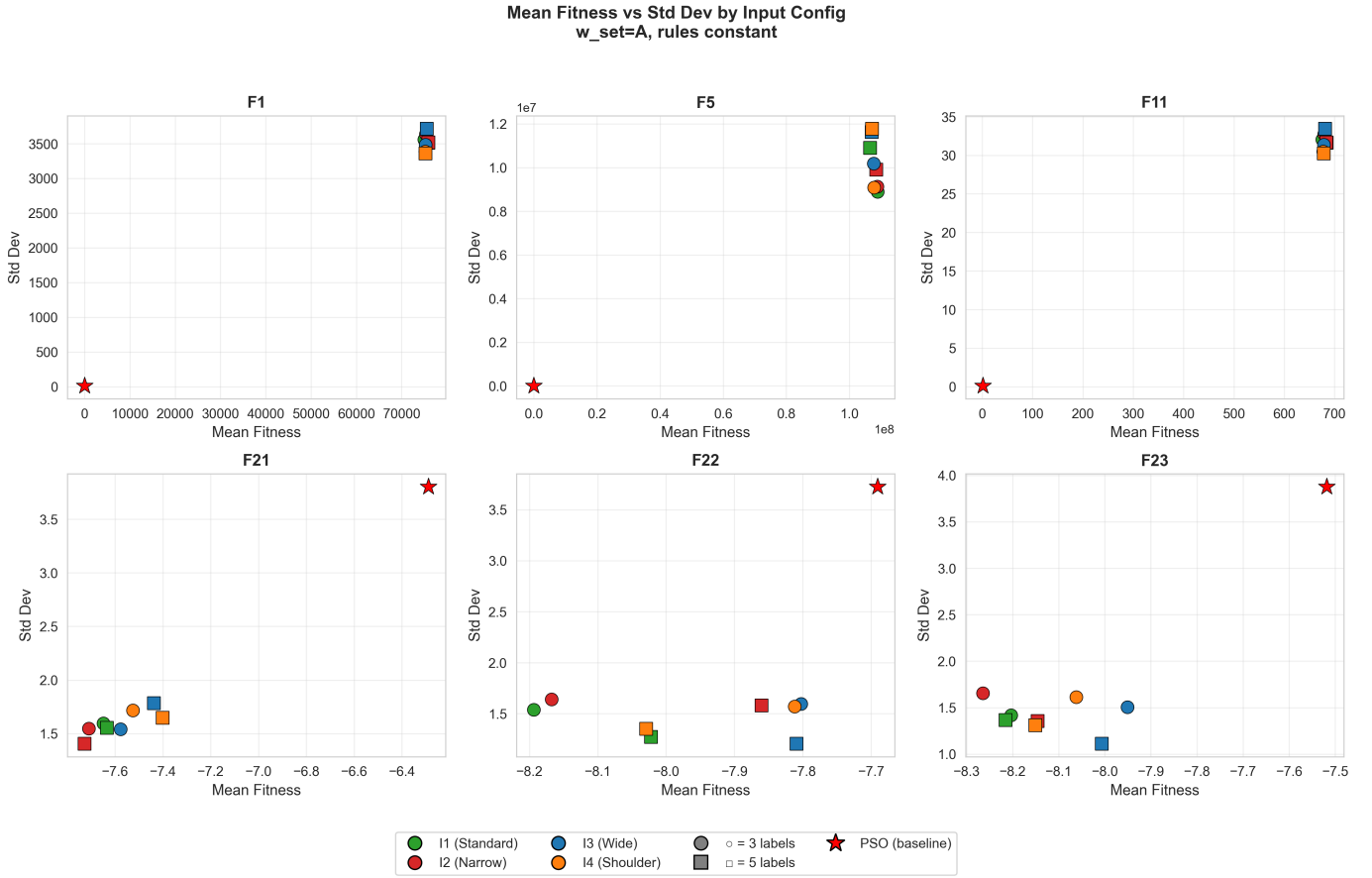


Fig. 3. Mean fitness vs. standard deviation for all configurations on each benchmark function. Colors encode input sets (I1–I4), markers distinguish granularity ($\circ$ =3L, $\square$ =5L), and the star denotes the PSO baseline. On multimodal functions, FCS variants achieve lower dispersion; on unimodal functions, all FCS variants are orders of magnitude worse than PSO.

TABLE VII

PSO VS BEST PSO–FCS VARIANT PER FUNCTION. FITNESS IMPROVEMENT, STANDARD DEVIATION REDUCTION, WILCOXON SIGNED-RANK p -VALUE, AND COMPUTATIONAL OVERHEAD ARE REPORTED OVER 31 INDEPENDENT RUNS.

Func.	Win	Best FCS	Impr.%	σ Red.%	p	OH%
F1	PSO	I1:3L	—	—	<0.001***	20.1
F5	PSO	I1:5L	—	—	<0.001***	27.6
F11	PSO	I1:3L	—	—	<0.001***	20.2
F21	FCS	I2:5L	+22.9	−63.0	0.041*	3.6
F22	FCS	I1:3L	+6.5	−58.7	0.824 ns	5.3
F23	FCS	I2:3L	+9.9	−57.3	0.504 ns	1.0

on F21 and F22.

C. Effect of Linguistic Granularity

The comparison between three-label and five-label partitions reveals a consistent advantage for the coarser granularity. As shown in Table VIII, the three-label configuration achieves a better mean rank than the five-label counterpart for three of the four input sets (I1, I2, I3), with the sole exception being I4 (Shoulder), where the five-label version slightly outperforms.

TABLE VIII

MEAN RANK OF EACH INPUT SET CONFIGURATION ACROSS SIX BENCHMARK FUNCTIONS (1 = BEST). RANKS COMPUTED INDEPENDENTLY PER GRANULARITY LEVEL.

Input Set	3-Label Rank	5-Label Rank
I1 (Standard)	2.83	4.50
I2 (Narrow)	3.67	5.17
I4 (Shoulder)	4.50	4.00
I3 (Wide)	5.50	5.83

The mean Gap% across all functions reinforces this finding: three-label configurations achieve an average gap of 23.7%, compared to 24.1% for five-label configurations. While the absolute difference is small, the direction is consistent, suggesting that five labels over-segment the input space without providing additional useful discrimination.

This result has a practical interpretation. The fuzzy controller operates on two normalized inputs (diversity and progress), each bounded in $[0, 1]$. With three labels, each linguistic term captures a broad region of the input space, leading to robust activation patterns where multiple rules contribute to the output. With five labels, the input space is

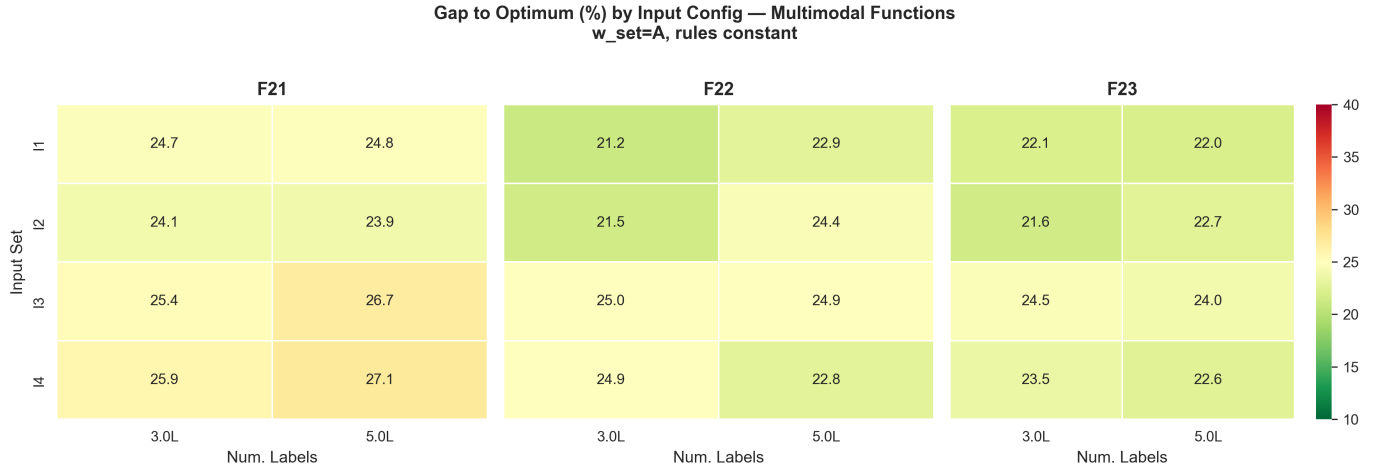


Fig. 4. Mean fitness heatmap for multimodal functions (F21–F23) across input sets and granularity levels. Darker cells indicate better fitness. I1 and I2 configurations consistently outperform I3 and I4.

partitioned more finely, but the underlying signals (population diversity and iteration count) do not exhibit the resolution needed to exploit this granularity. The result is a controller that responds to noise in the input measurements rather than to meaningful changes in search state.

VI. CONCLUSIONS

This work investigated the sensitivity of fuzzy-controlled PSO to the geometric design of input membership functions, a factor that has been consistently overlooked in prior research. By crossing four input set geometries (I1–I4) with two granularity levels (3 and 5 labels) under a constant output set and rule base, a total of 1,674 experiments were conducted across six benchmark functions with distinct landscape characteristics.

Three main findings emerge from the experimental analysis. First, the input membership function configuration has a measurable effect on controller performance. On the multimodal Shkel functions, the best fuzzy-controlled variant improved mean fitness by up to 22.9% and reduced standard deviation by up to 63% relative to the PSO baseline, while on unimodal functions the baseline PSO consistently dominated all fuzzy variants. This dichotomy confirms that diversity-driven inertia adaptation is beneficial only when the landscape contains deceptive local optima that require active exploration management.

Second, the geometric properties of the input partition determine the quality of the control signal. Configurations with moderate overlap (I1) achieved the best average rank (2.83 in the three-label case), while excessive overlap (I3) consistently produced the worst performance at both granularity levels. This result indicates that overly smooth partitions dilute the discrimination between distinct search states, leading to inertia adaptation that fails to react meaningfully to changes in diversity or progress.

Third, three-label partitions outperformed five-label partitions for three of the four input sets, with an average Gap% of

23.7% versus 24.1%. The coarser granularity produces more robust rule activation patterns on the two normalized inputs used in this controller, while five labels over-segment the input space without providing additional useful discrimination.

These findings have practical implications for the design of fuzzy-controlled metaheuristics. Input membership function geometry should be treated as an explicit design variable rather than a default choice, particularly when the controller operates on low-resolution signals such as normalized diversity and iteration progress. For practitioners, the recommendation emerging from this study is to favor moderate-overlap triangular partitions with three linguistic labels as a starting point for fuzzy inertia weight controllers.

Several limitations should be acknowledged. The study is restricted to triangular membership functions; other shapes (Gaussian, trapezoidal) may exhibit different sensitivity patterns. The benchmark suite, while structurally diverse, comprises only six functions and does not include high-dimensional multimodal problems where the interaction between geometry and granularity could differ. Additionally, the output set and rule base were held constant; their interaction with the input partition remains an open question.

Future work will extend this analysis in two directions: evaluating the interaction between input and output partition designs in a full factorial framework, and applying the methodology to combinatorial optimization problems where diversity dynamics differ substantially from continuous benchmarks.

ACKNOWLEDGMENT

José Barrera-García was supported by the National Agency for Research and Development ANID BECAS/DOCTORADO NACIONAL 21242516.

Felipe Cisternas-Caneo was supported by the National Agency for Research and Development ANID BECAS/DOCTORADO NACIONAL 21230203.

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